OAK INTERNATIONAL SCHOOL

Deliverable 3: Curriculum Alignment Matrix (IB Continuum vs. Beninese APC)

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Curricular Scope: All Seven Oak International School Classes (CM2/MYP 1 to Première/DP 2)

Standardised Alignment Taxonomy:

- **[DIR] Direct Alignment**: Substantially the same mathematical concept, comparable depth, and parallel cognitive expectations.
- **[PAR] Partial Alignment**: Common core concept, but differing in depth, pedagogical method, mathematical notation, or problem context.
- **[SEQ] Sequencing Mismatch**: Same concept taught at different chronological grades in the two programmes.
- **[IB-SPEC] IB-Specific Content**: Prescribed in the IB syllabus but absent from the corresponding Beninese national grade.
- **[APC-SPEC] APC-Specific Content**: Mandated by the Beninese national syllabus but absent from the corresponding IB level.
- **[GAP] Prerequisite Gap**: Concept required as assumed prior knowledge in one programme but not yet taught in the other.
- **[UNV] Unverified**: Insufficient official statutory evidence to establish a definitive curricular correspondence.

1. CLASS LEVEL 1: CM2 / MYP 1

Alignment ID	Mathematical Topic	Alignment Code	IB MYP 1 Expectation	Beninese APC CM2 Expectation
ALN-1.01	Whole Number Operations	[DIR] Direct Alignment	Standard algorithms for addition, subtraction, multiplication, and division of whole numbers up to millions.	Techniques opératoires complètes des quatre opérations sur les entiers naturels jusqu'au milliard; division euclidienne avec reste.
ALN-1.02	Decimal Operations & Metric Conversion	[DIR] Direct Alignment	Decimal place value to thousandths; adding and subtracting decimals; metric unit conversions (length, mass, capacity).	Nombres décimaux arithmétiques; calcul posé; conversions dans le système métrique (m, g, L) et unités agraires (ha, a, ca).
ALN-1.03	Fractions: Concept & Like Denominators	[DIR] Direct Alignment	Fractions as division/part-whole; equivalent fractions; adding and subtracting fractions with like/simple denominators.	Notion de fraction; simplification; égalité de fractions; somme et différence de fractions de même dénominateur ou dénominateurs multiples.
ALN-1.04	Commercial Arithmetic & Financial Maths	[PAR] Partial Alignment	Consumer maths: simple financial problems, discounts, budgeting, unitary cost.	Arithmétique commerciale structurée: prix d'achat, frais, prix de revient, prix de vente, bénéfice, perte, intérêt simple, capital.
ALN-1.05	Angles and Polygons	[DIR] Direct Alignment	Properties of triangles and quadrilaterals; angle types (acute, right, obtuse); angle measurement with protractor.	Classification des triangles et quadrilatères; reconnaissance et construction des angles droit, aigu, obtus, plat; médiatrice.
ALN-1.06	Perimeter and Area of Plane Figures	[DIR] Direct Alignment	Perimeter of polygons; area of rectangle, square, right triangle, and composite 2D shapes.	Formules de périmètre et d'aire du carré, rectangle, triangle, losange, trapèze et disque (\pi \approx 3,14).
ALN-1.07	Volume of Cubes & Rectangular Prisms	[PAR] Partial Alignment	Volume by counting unit cubes; formula V = l \times w \times h; informal capacity links.	Formule V = L \times l \times h; patron du pavé et du cube; équivalence métrique fondamentale: 1\text{ dm}^3 = 1\text{ L} = 1\text{ kg} d'eau pure.
ALN-1.08	Coordinate Geometry (1st Quadrant)	[IB-SPEC] IB-Specific	Plotting coordinates (x, y) in the first quadrant; reading line graphs and spatial grids.	Non prescrit au programme officiel du CM2 béninois (repérage purement intuitif sur quadrillage).
ALN-1.09	Probability Likelihood Scale	[IB-SPEC] IB-Specific	Likelihood scale (impossible, unlikely, even, likely, certain); expressing probability as simple fractions.	Non prescrit au cycle primaire béninois (introduit ultérieurement au second cycle).
ALN-1.10	Agrarian Units & Civil Calculations	[APC-SPEC] APC-Specific	Metric area only (\text{m}^2, \text{cm}^2, \text{km}^2). Non-metric/agrarian units not assessed.	Mesures agraires exigées au CEP béninois: hectare (ha), are (a), centiare (ca) et leurs relations avec le mètre carré.

Detailed Comparative Analysis (CM2 / MYP 1)

- **Similarities**: Common core mastery of elementary arithmetic, multi-digit operations, standard metric conversions, and perimeter/area calculations for standard polygons.
- **Depth Differences**: Beninese CM2 requires high calculation speed without calculators for compound operations, fractions, and multi-step commercial arithmetic problems. MYP 1 emphasizes conceptual representation, inquiry

into patterns (Criterion B), and verbal justifications.

- **Terminology & Notation**:
- English: *area*, *perimeter*, *simplifying fractions*, *decimal point* (3.14).
- French: *aire*, *périmètre*, *rendre irréductible une fraction*, *virgule décimale* (3,14).
- **Assessment Differences**: CEP tests structured step-by-step solutions (*Ce que je sais, ce que je cherche, solution, opérations, résultat*). MYP 1 evaluates against Criteria A, B, C, D with open-ended tasks and reflections.
- **Harmonisation Strategy**: Co-teach shared concepts (operations, fractions, geometric formulas); use dual-language terminology flashcards; conduct bilingual bridging tasks linking Beninese market calculations (*achats, ventes, bénéfice*) to MYP Criterion D real-life investigations.

2. CLASS LEVEL 2: SIXIÈME / MYP 2

Alignment ID	Mathematical Topic	Alignment Code	IB MYP 2 Expectation	Beninese APC 6ème Expectation
ALN-2.01	Divisibility & Prime Factorisation	[DIR] Direct Alignment	Divisibility tests; prime numbers; prime factor decomposition; finding HCF and LCM.	Critères de divisibilité (2, 3, 5, 9, 10); division euclidienne (dividende, diviseur, quotient, reste); décomposition élémentaire.
ALN-2.02	Fractions: Multiplication & Operations	[PAR] Partial Alignment	Multiplying and dividing fractions; mixed numbers; four operations with order of operations.	Fractions comme quotient de deux entiers; égalité de fractions; produit d'une fraction par un nombre; fractions décimales.
ALN-2.03	Axial Symmetry (Line Reflection)	[DIR] Direct Alignment	Reflections in line axes; properties of reflection (conservation of distance, angle, and shape).	(SA 5) Symétrie axiale / orthogonale par rapport à une droite: tracé, conservation des longueurs, angles, alignements et aires.
ALN-2.04	Synthetic Plane Geometry Foundations	[PAR] Partial Alignment	Identifying lines, rays, segments, parallel, and perpendicular lines; angle classification.	(SA 1) Notions ensemblistes de géométrie: appartenance (\in, \notin), droites sécantes, parallèles, perpendiculaires; cercle, corde, arc; médiatrice.
ALN-2.05	Solids: Cubes, Prisms & Nets	[DIR] Direct Alignment	Surface area and volume of right rectangular prisms and cubes; drawing 2D nets of 3D solids.	(SA 3) Pavé droit et cube: perspective cavalière, patrons, calculs d'aire latérale, aire totale et volume.
ALN-2.06	Directed Numbers (Negative Integers)	[SEQ] Sequencing Mismatch	Full introduction to negative numbers, number lines, addition and subtraction of directed numbers.	Reporté en 5ème au Bénin (SA 3). En 6ème béninoise, les calculs sont restreints aux entiers naturels et décimaux arithmétiques positifs.
ALN-2.07	Formal Geometric Proof Schema	[APC-SPEC] APC-Specific	Informal geometric reasoning and written justifications (Criterion C).	Enseignement obligatoire du schéma déductif rigoureux: « Données / On sait que... Propriété / Or si... Conclusion / Donc... ».
ALN-2.08	Four-Quadrant Cartesian Coordinates	[SEQ] Sequencing Mismatch	Full 4-quadrant coordinate plotting and geometric shape graphing.	Non prescrit en 6ème béninoise; repère cartésien introduit formellement en 5ème et 4ème.

Detailed Comparative Analysis (Sixième / MYP 2)

- **Similarities**: High convergence in synthetic planar geometry (segments, lines, angles, circles), axial symmetry, and basic 3D solids.
- **Depth Differences**: Beninese 6ème insists heavily on formal geometric deductive writing (*démonstration géométrique*) using axiomatic phrasing. MYP 2 pushes earlier into negative directed numbers and algebraic expressions.
- **Terminology & Notation**:
- English: *perpendicular bisector*, *line segment* [AB], *ray* [AB), *midpoint*.
- French: *médiatrice*, *segment* [AB], *demi-droite* [AB), *milieu*.
- **Sequencing Mismatch Strategy**: Deliver a 2-week bridging unit in MYP 2 on directed numbers to ensure students meet IB expectations without disrupting the Beninese positive-number progression.

3. CLASS LEVEL 3: CINQUIÈME / MYP 3

Alignment ID	Mathematical Topic	Alignment Code	IB MYP 3 Expectation	Beninese APC 5ème Expectation
ALN-3.01	Directed Numbers (\mathbb{Z} & Decimals)	[DIR] Direct Alignment	Operations with directed integers and decimals; brackets, signs, and order of operations.	(SA 3) Nombres relatifs (positifs et négatifs): comparaison, abscisse, addition, soustraction, suppression des parenthèses.
ALN-3.02	Fractions: Addition with Unlike Denominators	[DIR] Direct Alignment	Adding and subtracting fractions with different denominators; multiplying and dividing fractions.	Addition et soustraction de fractions de dénominateurs distincts (recherche du dénominateur commun); multiplication de fractions.
ALN-3.03	Parallel Lines and Angles	[DIR] Direct Alignment	Angle properties of parallel lines cut by a transversal: alternate interior, corresponding, co-interior.	(SA 2) Angles formés par deux droites parallèles et une sécante: alternes-internes, alternes-externes, correspondants.
ALN-3.04	Triangle Angles & Special Lines	[DIR] Direct Alignment	Triangle angle sum (180^\circ); exterior angle theorem; mediators, angle bisectors, altitudes.	Somme des angles d'un triangle (180^\circ); inégalité triangulaire; médiatrices (cercle circonscrit) et bissectrices (cercle inscrit).
ALN-3.05	Central Symmetry (Point Reflection)	[DIR] Direct Alignment	Rotations of 180^\circ around a point; properties of point reflection; rotational symmetry.	(SA 4) Symétrie centrale (par rapport à un point): définition, constructions, propriétés de conservation, lien avec le parallélogramme.
ALN-3.06	Parallelogram Properties	[DIR] Direct Alignment	Properties and tests for parallelograms, rhombuses, rectangles, and squares.	Définition du parallélogramme; caractérisation par ses diagonales, ses côtés opposés et son centre de symétrie.
ALN-3.07	Powers of 10 & Scientific Notation	[DIR] Direct Alignment	Scientific notation a \times 10^k; integer powers; estimation and orders of magnitude.	Puissances de 10 à exposants entiers relatifs; notation scientifique; ordre de grandeur.
ALN-3.08	Proportionality & Linear Applications	[DIR] Direct Alignment	Direct proportion; constant of proportionality; conversion graphs; speed, distance, time.	(SA 5) Quatrième proportionnelle; coefficient de proportionnalité; vitesse moyenne; pourcentages; échelles.
ALN-3.09	Two-Stage Probability & Tree Diagrams	[IB-SPEC] IB-Specific	Sample space diagrams, two-way tables, probability tree diagrams for independent events.	Non prescrit au premier cycle secondaire béninois (absent du programme de 5ème).
ALN-3.10	Linear Sequences & n^{\text{th}} Term Rule	[IB-SPEC] IB-Specific	Investigating linear and non-linear patterns, expressing n^{\text{th}} term algebraically (Criterion B).	Non prescrit en 5ème béninoise (les suites sont abordées au second cycle).

Detailed Comparative Analysis (Cinquième / MYP 3)

- **Direct Matches**: Outstanding congruence across geometry (parallel angles, triangle angle sum, central symmetry, parallelograms) and arithmetic (relative numbers, fraction operations, powers of 10, and proportionality).
- **Instructional Coordination**: The two programmes run almost in lockstep for geometry and signed numbers during this academic year.

- **Harmonisation Strategy**: Teach the shared core geometry and rational numbers concurrently; allocate dedicated MYP inquiry sessions for probability tree diagrams and sequence pattern investigations.

4. CLASS LEVEL 4: QUATRIÈME / MYP 4

Alignment ID	Mathematical Topic	Alignment Code	IB MYP 4 Expectation	Beninese APC 4ème Expectation
ALN-4.01	Pythagoras' Theorem & Converse	[DIR] Direct Alignment	Applying Pythagoras' theorem in 2D and 3D; converse of Pythagoras to prove right angles.	(SA 1) Théorème de Pythagore et sa réciproque dans le triangle rectangle; applications numériques et démonstrations.
ALN-4.02	Right Triangle Trigonometry (Cosine)	[PAR] Partial Alignment	Full introduction to \sin, \cos, and \tan in right-angled triangles; finding sides and angles.	Cosinus d'un angle aigu dans le triangle rectangle (\cos = \text{adjacent}/\text{ hypoténuse}); sinus et tangente reportés en 3ème.
ALN-4.03	Rational Numbers (\mathbb{Q}) & Powers	[DIR] Direct Alignment	Arithmetic with rational numbers; rules of integer exponents (a^m \times a^n, (a^m)^n, a^{-n}).	(SA 2) Nombres rationnels: opérations, signes, fractions irréductibles; puissances à exposants entiers relatifs.
ALN-4.04	Algebraic Expansion & Simple Factoring	[DIR] Direct Alignment	Expanding single and double brackets; factoring by common factors; simplifying expressions.	Calcul littéral: réduction, développement par simple et double distributivité; factorisation élémentaire.
ALN-4.05	Linear Equations in One Unknown	[DIR] Direct Alignment	Solving linear equations ax + b = cx + d with integer and fractional coefficients.	Résolution d'équations du premier degré à une inconnue dans \mathbb{Q}; problèmes à mettre en équation.
ALN-4.06	Vector Concept & Translation	[PAR] Partial Alignment	Column vectors \begin{pmatrix} x \\ y \end{pmatrix}; translation vectors; vector addition.	(SA 4) Notion géométrique de vecteur: direction, sens, norme; égalité vectorielle; relation de Chasles; translation.
ALN-4.07	Midpoint Theorem & Triangle Concurrency	[APC-SPEC] APC-Specific	Midpoints used in coordinate geometry; informal triangle centres.	(SA 1) Théorème des milieux dans le triangle; concourance des 4 droites remarquables (orthocentre, centre de gravité G).
ALN-4.08	Pyramids and Cones: Volume & Surface	[DIR] Direct Alignment	Surface area and volume of regular pyramids and right cones (V = \frac{1}{3}Bh).	(SA 3) Pyramide régulière et cône de révolution: perspective cavalière, patrons, calculs de volume et aire latérale.
ALN-4.09	Cumulative Frequency & Boxplots	[IB-SPEC] IB-Specific	Cumulative frequency curves, quartiles, IQR, box-and-whisker plots, scatter plots.	Effectifs et fréquences cumulés prescrits, mais boxplots et quartiles non abordés en 4ème béninoise (reportés en 2nde).

Detailed Comparative Analysis (Quatrième / MYP 4)

- **Similarities**: Complete shared commitment to Pythagoras' theorem, rational number arithmetic, algebraic expansion, and right pyramid/cone mensuration.
- **Divergence**:
- Trigonometry: MYP 4 teaches all three ratios (\sin, \cos, \tan); Beninese 4ème teaches strictly the cosine (\cos), reserving sine and tangent for 3ème.

- Vectors: Beninese 4ème establishes a rigorous synthetic vector framework (direction, sense, norm, Chasles' relation \vec{AB}+\vec{BC}=\vec{AC}), providing an exceptional geometric foundation that enriches the IB algebraic vector perspective.
- **Harmonisation Strategy**: Introduce \sin and \tan as natural extensions of \cos during the trigonometry unit; leverage the Beninese geometric vector foundation to provide intuitive depth for MYP column vectors.

5. CLASS LEVEL 5: TROISIÈME / MYP 5

Alignment ID	Mathematical Topic	Alignment Code	IB MYP 5 Expectation	Beninese APC 3ème Expectation
ALN-5.01	Thales' Theorem & Similarity	[DIR] Direct Alignment	Intercept theorem / Thales' theorem; properties of similar triangles and scale factors (k, k^2, k^3).	(SA 1) Propriété de Thalès et sa réciproque dans le triangle; calcul de longueurs; agrandissement et réduction.
ALN-5.02	Right-Angled Trigonometry & Identities	[DIR] Direct Alignment	\sin, \cos, \tan; exact values (30^\circ, 45^\circ, 60^\circ); identities \sin^2\theta + \cos^2\theta = 1, \tan\theta = \frac{\sin\theta}{\cos\theta}.	Trigonométrie dans le triangle rectangle; lignes trigonométriques; relations fondamentales; angles remarquables.
ALN-5.03	Real Numbers, Radicals & Surds	[DIR] Direct Alignment	Real numbers (\mathbb{R}); simplifying radicals \sqrt{ab}=\sqrt{a}\sqrt{b}; rationalising denominators with conjugates.	(SA 2) Nombres réels; calcul sur les radicaux; simplification \sqrt{a^2b}=a\sqrt{b}; expression conjuguée au dénominateur.
ALN-5.04	Algebraic Identities & Factorisation	[DIR] Direct Alignment	Expanding and factoring using identities: (a\pm b)^2, a^2-b^2; factoring quadratic expressions.	Produits remarquables; factorisation complète de polynômes; fractions rationnelles (valeurs interdites, simplification).
ALN-5.05	Linear Systems of Equations (2 \times 2)	[DIR] Direct Alignment	Solving simultaneous linear equations in 2 variables algebraically and graphically.	Systèmes de deux équations du premier degré à deux inconnues: substitution, combinaison, interprétation graphique.
ALN-5.06	Null-Factor Equations & Quadratic Roots	[PAR] Partial Alignment	Solving quadratic equations by factoring, completing the square, and quadratic formula.	Équations-produits (ax+b)(cx+d)=0; factorisation pour trouver les racines; la formule générale du discriminant \Delta est formellement réservée à la Seconde.
ALN-5.07	Coordinate Geometry & Linear Functions	[DIR] Direct Alignment	Equation of line y = mx + c; midpoint, distance formula; parallel and perpendicular line criteria.	(SA 3 & 5) Repère orthonormé; coordonnées d'un vecteur; distance AB; colinéarité, orthogonalité; fonctions affines f(x)=ax+b.
ALN-5.08	Inscribed Angles & Circle Geometry	[DIR] Direct Alignment	Angle at centre is twice angle at circumference; angles in the same segment; cyclic quadrilaterals.	(SA 1) Angles inscrits et angles au centre interceptant le même arc dans un cercle; polygones réguliers.
ALN-5.09	Grouped Continuous Data & Histograms	[DIR] Direct Alignment	Histograms, modal class, estimating mean from grouped continuous data.	(SA 5) Séries statistiques groupées en classes d'amplitudes; histogrammes; moyenne pondérée; classe modale.
ALN-5.10	Sine & Cosine Rules (Non-Right Triangles)	[IB-SPEC] IB-Specific	Sine rule (\frac{a}{\sin A} = \frac{b}{\sin B}) and cosine rule (a^2 = b^2+c^2-2bc\cos A) in general triangles.	Non prescrit en 3ème béninoise (Al-Kashi est enseigné en Seconde C/D).
ALN-5.11	High-Stakes National Exam Rubric	[APC-SPEC] APC-Specific	MYP Criteria A, B, C, D assessed on 0–8 rubrics.	Exigence statutaire du BEPC: situation d'évaluation à 3 problèmes notés sur les critères C_a, C_m, C_o et CP.

Detailed Comparative Analysis (Troisième / MYP 5)

- **High Strategic Importance**: This is a double-graduation year (BEPC national diploma and MYP 5 completion / eAssessment).
- **Curricular Convergence**: The academic overlap is remarkably strong (~85%), spanning Thales' theorem, trigonometry, coordinate geometry, linear systems, affine functions, circle theorems, and radicals.
- **Harmonisation Strategy**: Run a harmonised core curriculum through Terms 1 and 2; in Term 3, provide targeted bifurcation:
- Track A: BEPC examination workshops emphasizing 3-problem contextual scenarios, criterion rubrics (C_a, C_m, C_o), and non-calculator paper-and-pencil rigor;
- Track B: MYP 5 inquiry tasks, Criterion B pattern investigations, and general triangle trigonometry (sine/cosine rules).

6. CLASS LEVEL 6: SECONDE / DP 1

Alignment ID	Mathematical Topic	Alignment Code	IB DP 1 Expectation (Common Core / AA)	Beninese APC 2nde C/D Expectation				
ALN-6.01	Subsets of \mathbb{R}, Intervals & Absolute Value	[PAR] Partial Alignment	Real numbers, interval notation, modulus function	x	, solving	ax+b	=c.	Sous-ensembles \mathbb{N}\subset\mathbb{Z}\subset\mathbb{D}\subset\mathbb{Q}\subset\mathbb{R}; intervalles; valeur absolue comme distance; encadrements, majorants/minorants.
ALN-6.02	Quadratic Theory & Discriminant \Delta	[DIR] Direct Alignment	Quadratic functions: vertex form, standard form, discriminant \Delta = b^2-4ac, roots, sign of quadratic.	Trinôme du second degré: forme canonique, calcul du discriminant \Delta, factorisation, somme et produit (S, P), tableaux de signes.				
ALN-6.03	Concept of Function & Reference Functions	[DIR] Direct Alignment	Function concept, domain, range, graph, composite functions, transformations (f(x)+k, f(x+k), -f(x)).	Notion d'application et de fonction, ensemble de définition D_f, parité, périodicité, monotonie, fonctions de référence (x^2, \frac{1}{x}, \sqrt{x},	x	).		
ALN-6.04	Scalar Product (Dot Product) in 2D	[DIR] Direct Alignment	Dot product \vec{u}\cdot\vec{v} =	\vec{u}		\vec{v}	\cos\theta = u_1v_1 + u_2v_2; angle between vectors; perpendicularity.	Produit scalaire de deux vecteurs: définition géométrique, projection orthogonale, forme analytique (xx'+yy'), orthogonalité.

Alignment ID	Mathematical Topic	Alignment Code	IB DP 1 Expectation (Common Core / AA)	Beninese APC 2nde C/D Expectation				
ALN-6.05	Law of Cosines (Al-Kashi)	[DIR] Direct Alignment	Cosine rule: a^2 = b^2 + c^2 - 2bc \cos A; applications to triangle solving.	Relations métriques dans le triangle: théorème d'Al-Kashi (a^2=b^2+c^2-2bc\cos\hat{A}); théorème de la médiane.				
ALN-6.06	Unit Circle & Radian Trigonometry	[DIR] Direct Alignment	Radian measure (\pi = 180^\circ); unit circle; \cos, \sin, \tan; exact values; double-angle formulas (\sin 2\theta, \cos 2\theta).	Cercle trigonométrique, enroulement de la droite numérique, radians, angles associés, formules d'addition et de duplication.				
ALN-6.07	Sequences & Financial Mathematics	[SEQ] Sequencing Mismatch	Arithmetic and geometric sequences; sum formulas; compound interest, depreciation, loans.	Reporté en Première dans le programme béninois (absent du programme officiel de Seconde).				
ALN-6.08	Incidence Geometry of 3D Space	[APC-SPEC] APC-Specific	Informal 3D coordinates and solid geometry.	Axiomes d'incidence de l'espace; théorèmes de parallélisme et d'orthogonalité entre droites et plans dans l'espace.				
ALN-6.09	Logarithms & Exponential Equations	[SEQ] Sequencing Mismatch	Logarithm laws (\log(xy), \log(x/y), \ln x); solving exponential equations a^x = b.	Reporté en Terminale dans le programme béninois (fonctions \ln et \exp).				
ALN-6.10	Graphic Display Calculator (GDC)	[IB-SPEC] IB-Specific	Pervasive use of GDC for graphing, numerical roots, and statistical distributions.	Interdit ou inexistant dans les examens béninois (calculatrices scientifiques simples non programmables uniquement).				

Detailed Comparative Analysis (Seconde / DP 1)

- **Mathematical Convergence**: The core pillars of Seconde C/D—quadratic equations, functional analysis, 2D vector dot products, Al-Kashi theorem, and unit-circle radian trigonometry—match DP Topic 2 (Functions), Topic 3 (Geometry & Trig), and Topic 1 (Algebra).
- **Sequencing Discrepancies**: DP 1 introduces sequences and logarithms early; Beninese Seconde postpones sequences to Première and logarithms to Terminale, but delves significantly deeper into pure synthetic 3D incidence geometry and formal intervals/inequalities.
- **Harmonisation Strategy**: Co-teach quadratics, functions, trigonometry, and dot products as an integrated master course. Schedule dedicated DP 1 modules for sequences and logarithmic foundations, while preserving the Beninese rigorous proofs in 3D geometry and metric properties.

7. CLASS LEVEL 7: PREMIÈRE / DP 2

Alignment ID	Mathematical Topic	Alignment Code	IB DP 2 Expectation (Common Core / AA)	Beninese APC 1ère C/D Expectation
ALN-7.01	Differential Calculus: Derivative & Tangent	[DIR] Direct Alignment	Derivative as instantaneous rate of change; derivative of power, rational, trig functions; equation of tangent; optimization.	Taux de variation; nombre dérivé comme limite; coefficient directeur de la tangente; calcul des dérivées; sens de variation et extremums.
ALN-7.02	Limits, Asymptotes & Curve Sketching	[DIR] Direct Alignment	Limits at infinity; vertical and horizontal asymptotes; curve sketching; finding local extrema.	Calcul de limites en un point et à l'infini; formes indéterminées; asymptotes verticales, horizontales, obliques; plan complet d'étude.
ALN-7.03	Sequences: Arithmetic & Geometric	[DIR] Direct Alignment	Arithmetic and geometric sequences: explicit formulas, recursive rules, sum of n terms.	Suites arithmétiques et géométriques: définition récurrente, terme général, somme de termes consécutifs; monotonie, bornes.
ALN-7.04	Combinatorics & Probability Distributions	[DIR] Direct Alignment	Permutations, combinations \binom{n}{r}, binomial theorem; discrete random variables, expectation E(X), variance, binomial distribution.	Dénombrement (p-listes, A_n^p, C_n^p); triangle de Pascal, binôme de Newton; probabilités conditionnelles; variables aléatoires, espérance, variance.
ALN-7.05	Advanced Trigonometric Equations	[DIR] Direct Alignment	Solving \cos x = \cos a, \sin x = \sin a; compound and double-angle formulas.	Résolution d'équations trigonométriques dans \mathbb{R}; formules de transformation de produits et sommes (formules de Simpson).
ALN-7.06	Barycentres (Weighted Averages of Points)	[APC-SPEC] APC-Specific	Concept of center of mass used informally in physics/mechanics, but not part of the DP mathematics syllabus.	Barycentre de 2, 3, 4 points pondérés; associativité (barycentres partiels); réduction vectorielle; lignes de niveau (outil majeur au Bac).
ALN-7.07	Integral Calculus & Areas	[SEQ] Sequencing Mismatch	Anti-derivatives; definite integrals \int_a^b f(x)\,dx; Fundamental Theorem of Calculus; area under curves.	Reporté en classe de Terminale dans le programme national béninois (où le calcul intégral est enseigné de façon exhaustive).
ALN-7.08	Linear Programming via Half-Planes	[APC-SPEC] APC-Specific	Addressed in DP AI, but absent from DP AA syllabus.	Systèmes d'inéquations linéaires à deux inconnues; détermination de polygones de contraintes; optimisation de fonction économique au programme de 1ère.
ALN-7.09	DP Internal Assessment (Exploration)	[IB-SPEC] IB-Specific	12–20 page individual mathematical exploration demonstrating personal engagement, communication, and reflection.	Non existant dans le système national béninois (évaluation fondée exclusivement sur des épreuves écrites ponctuelles).

Detailed Comparative Analysis (Première / DP 2)

- **High Mathematical Synergies**: Exceptional convergence in differential calculus, curve sketching, numerical sequences, and combinatorics/probability distributions.

- **Specific Structural Mismatches**:
- *Barycentres*: A massive, highly examined topic in Beninese 1ère C/D that does not appear in DP. This requires dedicated instructional time to guarantee success on national tests.
- *Integration*: DP 2 mandates integral calculus for area calculations and the Fundamental Theorem of Calculus; Beninese students do not officially study integrals until Terminale.
- **Harmonisation Strategy**: Deliver an intensive shared semester on differential calculus, polynomial/rational function analysis, numerical sequences, and combinatorics. Dedicate targeted parallel tracks for:
- Track A (Benin National Preparation): Barycentres, partial barycentres proofs, level curves, and linear programming;
- Track B (IB DP Preparation): Introduction to anti-differentiation, definite integrals, and coaching on the Internal Assessment Exploration.

End of Deliverable 3 — Curriculum Alignment Matrix.